

People Analytics II – Written Report

TechSolutions Inc.

Strategic Attrition Analysis and Retention Recommendations

Talent Development and Human Resources

Group 3:

Amandil Fidaleo, Catalina Sánchez, Marta Leitão, Marco Celoria, Stefan Dragan

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Executive Summary

Employee attrition represents one of the most significant workforce challenges facing organizations today. Beyond replacement costs, employee departures disrupt productivity, increase workload pressures on remaining teams, and weaken long-term talent pipelines.

At TechSolutions, overall attrition stands at 16.1%. While this figure appears manageable at first glance, a deeper analysis reveals that turnover is concentrated within a limited number of employee segments rather than being evenly distributed across the workforce.

Three findings emerged from the analysis.

First, attrition is heavily concentrated among frontline and early-career employees. Sales Representatives experience attrition of 39.8%, while entry-level employees leave at a rate of 26.3%. The risk increases substantially when multiple factors coexist. Sales Representatives working overtime experience attrition of 66.7%, more than four times the organizational average.

Second, employee departures are consistently associated with workload pressure, compensation levels, ownership incentives, and engagement. Employees working overtime leave at nearly three times the rate of employees without overtime exposure. Employees in the lowest salary quartile experience attrition of 29.3%, while employees receiving stock ownership incentives demonstrate significantly stronger retention outcomes. Similarly, employees reporting the lowest levels of job involvement leave at substantially higher rates than highly engaged employees.

Third, predictive modelling confirms these patterns. A Random Forest classification model identified monthly income, age, tenure, overtime exposure, and stock option eligibility as the strongest predictors of attrition risk. The model achieved an AUC of 0.73, demonstrating a good ability to distinguish employees who leave from those who remain.

The findings indicate that TechSolutions does not face a company-wide retention problem. Instead, turnover is concentrated among identifiable employee populations and driven by a consistent set of workforce factors. As a result, targeted interventions focused on workload management, compensation competitiveness, ownership incentives, and manager-led engagement initiatives are likely to generate greater impact than broad organization-wide retention programs.

Business Context

TechSolutions employs 1,470 employees across Sales, Research & Development, and Human Resources functions. During the period analysed, 237 employees left the organisation, resulting in an overall attrition rate of 16.1%.

Workforce Overview

- Total Employees: 1,470
- Employees Retained: 1,233
- Employees Who Left: 237
- Attrition Rate: 16.1%

While the overall attrition rate provides a useful starting point, it offers limited insight into where turnover occurs and which employee groups are most affected. Understanding these patterns is particularly important because employee departures rarely affect organisations evenly. Attrition tends to concentrate within specific functions, career stages, and workforce segments, creating localized talent shortages and operational disruption.

For TechSolutions, the central business question is therefore not whether attrition exists, but where it occurs, what drives it, and how management can most effectively intervene.

Analytical Approach

The analysis followed a three-stage People Analytics framework designed to move from observation to action.

1. The first stage focused on descriptive analysis. Workforce data was examined to identify which employee groups experienced the highest levels of attrition. This included analysis of demographic characteristics, job roles, career levels, compensation, engagement indicators, and work conditions.
2. The second stage focused on diagnostic analysis. Relationships between workforce variables and attrition outcomes were assessed to identify the factors most strongly associated with employee departures.
3. The third stage focused on predictive analysis. A Random Forest classification model was developed to assess whether historical workforce patterns could be used to identify employees at elevated risk of future attrition. In addition to estimating attrition risk, the model provided insight into the relative importance of different workforce variables.

This combined approach allowed the analysis to answer three critical questions:

- Who is leaving?
- Why are they leaving?
- Which employees are most likely to leave in the future?

Workforce Attrition Patterns

Attrition Is Concentrated Among Early-Career Employees

Attrition varies considerably across age groups. Employees aged 18–25 experience attrition of 35.8%, more than double the organisational average. Attrition declines throughout mid-career stages before increasing slightly among employees approaching retirement age.

This pattern highlights a challenge common to many organisations: retaining employees during the earliest stages of their careers. Early-career turnover reduces the return on recruitment, onboarding, and training investments while creating ongoing pressure on talent acquisition teams.

The findings suggest that improving retention among younger employees should be viewed as a strategic workforce priority rather than a purely operational concern.

Frontline Sales Roles Represent the Highest Retention Risk

Attrition rates also vary significantly across job roles.

Sales Representatives experience attrition of 39.8%, making them the highest-risk employee group in the organisation. Laboratory Technicians (23.9%) and Human Resources employees (23.1%) also experience elevated turnover levels. At the other end of the spectrum, Managers (4.9%) and Research Directors (2.5%) demonstrate strong retention outcomes.

The concentration of turnover within customer-facing and operational roles creates risks that extend beyond replacement costs. Frequent departures disrupt customer relationships, increase onboarding requirements, and place additional workload pressure on remaining employees.

As a result, frontline retention represents both an HR challenge and a business performance issue.

Drivers of Attrition

While attrition is concentrated among specific employee groups, understanding the underlying drivers is essential for identifying effective interventions. Four factors consistently emerged

throughout the analysis: overtime exposure, compensation, ownership incentives, and employee engagement.

Overtime Is the Strongest Operational Driver

Among all variables analysed, overtime displayed the clearest relationship with employee attrition.

Employees working overtime experience attrition of 30.5%, compared with 10.4% among employees who do not work overtime. More than half of employees who left the organisation reported overtime exposure, while less than one quarter of retained employees worked overtime.

The relationship becomes even stronger when overtime is combined with other risk factors. Entry-level employees working overtime experience attrition of 52.6%, while Sales Representatives working overtime reach 66.7%.

These findings suggest that workload pressure is not simply a productivity concern. In several high-risk populations, overtime appears closely linked to turnover outcomes. Managing workload distribution and staffing capacity therefore represents one of the most direct opportunities to improve retention.

Compensation Plays a Significant Role in Retention

Compensation differences between employees who leave and those who remain are substantial.

Employees who stayed with the organisation earned an average monthly income of \$6,833. Employees who left earned an average of \$4,787, representing a difference of approximately 30%.

Further analysis reinforces this relationship. Employees in the lowest income quartile experience attrition of 29.3%, while employees in the highest quartile experience attrition of 10.3%.

The pattern is consistent across multiple analyses. Employees receiving lower levels of financial reward are more likely to leave, particularly when combined with overtime exposure and limited career progression opportunities.

While compensation is rarely the sole reason employees leave, it clearly influences retention outcomes within the populations most affected by attrition.

Ownership Incentives Strengthen Employee Retention

Stock ownership appears closely associated with employee retention.

Employees without stock option eligibility experience attrition of 24.4%. By comparison, employees receiving Stock Option Level 2 experience attrition of only 7.6%.

Ownership incentives create a longer-term relationship between employees and organisational performance. The findings suggest that employees with a stronger financial stake in company success are more likely to remain with the organisation.

This effect is particularly relevant for frontline and early-career employees, where attrition rates are highest and opportunities for ownership participation are often limited.

Engagement and Workplace Experience Matter

Compensation and workload explain only part of the attrition challenge.

Employees reporting the lowest level of job involvement experience attrition of 33.7%, compared with only 9.0% among employees reporting the highest level of involvement.

A similar pattern emerges for environmental satisfaction. Employees with the lowest satisfaction scores leave at nearly twice the rate of employees with higher satisfaction levels.

These findings suggest that employee retention is influenced not only by what employees receive from the organisation, but also by how they experience their work environment. Employees who feel connected to their work, supported by their managers, and engaged in their roles demonstrate substantially stronger retention outcomes.

Predictive Analysis

Building the Attrition Prediction Model

The descriptive and diagnostic analyses identified several factors consistently associated with employee departures. To determine whether these patterns could be used to identify future attrition risk, a predictive model was developed.

A Random Forest classification model was selected because it can analyse complex relationships among workforce variables and identify interactions that may not be visible through traditional statistical methods.

The model incorporated 29 workforce variables spanning four dimensions:

- Demographics

- Compensation and rewards
- Career progression
- Work conditions and engagement

The dataset was divided into training and testing samples.

- Training Sample: 1,176 employees (80%)
- Testing Sample: 294 employees (20%)

To improve reliability, the model was evaluated using five-fold cross-validation and class-balanced weighting techniques.

Model Performance

The model achieved the following performance metrics:

- Accuracy: 80.3%
- AUC-ROC: 0.73
- Cross-Validated AUC: 0.748 ± 0.013
- Recall: 19.1%

Given that attrition affects only 16.1% of employees, AUC provides the most informative measure of model quality. An AUC of 0.73 indicates that the model can distinguish employees who leave from employees who stay approximately 73% of the time.

The consistency of the cross-validation results suggests that the model performs reliably across different employee samples and is unlikely to be dependent on a single subset of data.

While the model is not intended to predict individual departures with certainty, it provides a practical mechanism for identifying elevated attrition risk and prioritising retention efforts.

Key Predictors of Attrition

The model identified ten variables with the greatest predictive importance.

Top predictors included:

- Monthly Income
- Age
- Total Working Years
- Overtime Exposure
- Years at Company
- Daily Rate

- Years with Current Manager
- Stock Option Level
- Distance from Home
- Number of Previous Employers

Several observations emerge from these results.

- First, compensation variables consistently rank among the strongest predictors of attrition.
- Second, career progression and tenure-related variables play a significant role in retention outcomes.
- Third, overtime remains one of the most influential operational drivers despite the presence of compensation and tenure variables.
- Most importantly, the predictive model reinforces the findings observed throughout the descriptive analysis. The same factors associated with historical attrition also emerge as the strongest indicators of future turnover risk.

This consistency increases confidence that the identified drivers represent meaningful workforce patterns rather than isolated statistical relationships.

Strategic Recommendations

The findings indicate that broad organisation-wide retention initiatives are unlikely to generate the greatest return on investment. Attrition is concentrated among identifiable employee populations and driven by a relatively small number of workforce factors.

Three strategic priorities should therefore guide retention efforts.

Recommendation 1: Reduce Overtime Exposure in Frontline Roles

Overtime represents the strongest operational driver of attrition and is particularly prevalent among high-risk employee populations.

TechSolutions should prioritise workload management initiatives within Sales and other overtime-intensive functions. Recommended actions include workload monitoring, staffing reviews, manager accountability measures, and improved workforce planning.

Priority Population:

- Sales Representatives
- Entry-Level Employees

- Overtime-Exposed Teams

Expected Outcomes:

- Reduced burnout risk
- Lower frontline attrition
- Improved workforce sustainability

Recommendation 2: Strengthen Compensation and Ownership Incentives

Compensation and ownership incentives emerged as consistent predictors of retention outcomes.

TechSolutions should conduct targeted compensation reviews focused on lower-income employee groups and critical frontline roles. In parallel, stock ownership opportunities should be expanded to a broader segment of early-career employees.

Priority Population:

- Lowest Income Quartile Employees
- Entry-Level Employees
- Sales Representatives

Expected Outcomes:

- Improved retention among high-risk employee groups
- Greater organisational commitment
- Reduced voluntary turnover

Recommendation 3: Increase Engagement Through Manager-Led Interventions

Engagement-related factors consistently differentiated employees who stayed from those who left.

Rather than implementing broad engagement programmes, TechSolutions should focus on manager capability, career development visibility, onboarding quality, and employee listening initiatives within high-risk teams.

Priority Population:

- Sales Teams
- New Hires
- Early-Career Employees

- Employees Reporting Low Engagement Levels

Expected Outcomes:

- Increased job involvement
- Improved workplace satisfaction
- Stronger organisational attachment

7. Conclusion

The objective of this analysis was to identify where attrition occurs, understand the factors associated with employee departures, and determine where intervention efforts are most likely to generate measurable impact.

Across descriptive, diagnostic, and predictive analyses, a consistent picture emerged.

Attrition is concentrated among early-career and frontline employees, particularly Sales Representatives and employees exposed to overtime. Compensation, ownership incentives, engagement, and career progression factors consistently differentiate employees who leave from employees who remain.

The predictive model validated these findings, identifying compensation, age, tenure, and overtime exposure as the strongest predictors of future attrition risk.

The evidence points toward a focused workforce challenge rather than an organisation-wide retention issue. As a result, retention efforts should prioritise the employee populations where turnover risk is highest and where interventions can generate the greatest impact.

By addressing workload pressures, improving compensation competitiveness, expanding ownership opportunities, and strengthening employee engagement, TechSolutions can reduce avoidable turnover, improve workforce stability, and protect long-term organisational performance.

Appendix 1: Workforce Overview

Metric	Value
Total Employees	1,470
Employees Retained	1,233
Employees Who Left	237
Overall Attrition Rate	16.1%

Appendix 2: Attrition by Age Group

Age Group	Attrition Rate
18–25	35.8%
26–35	19.1%
36–45	9.2%
46–55	11.5%
56+	17.0%

Appendix 3: Attrition by Job Role

Job Role	Attrition Rate
Sales Representative	39.8%
Laboratory Technician	23.9%
Human Resources	23.1%
Sales Executive	17.5%
Research Scientist	16.1%
Healthcare Representative	6.9%
Manufacturing Director	6.9%
Manager	4.9%
Research Director	2.5%

Appendix 4: Overtime Analysis

Group	Attrition Rate
Overtime	30.5%
No Overtime	10.4%

Employee Group	Overtime Exposure
Employees Who Left	53.6%

Employee Group	Overtime Exposure
Employees Who Stayed	23.4%

Appendix 5: Compensation Analysis

Metric	Value
Average Monthly Income (Stayers)	\$6,833
Average Monthly Income (Leavers)	\$4,787
Income Gap	30%

Appendix 6: Salary Quartiles

Salary Quartile	Attrition Rate
Q1 (<\$2,911)	29.3%
Q2 (\$2,911–\$4,919)	14.2%
Q3 (\$4,919–\$8,379)	10.6%
Q4 (>\$8,379)	10.3%

Appendix 7: Stock Options

Stock Option Level	Attrition Rate
Level 0	24.4%
Level 1	9.4%
Level 2	7.6%
Level 3	17.6%

Appendix 8: Engagement Metrics

Job Involvement Level	Attrition Rate
Level 1	33.7%
Level 2	18.9%
Level 3	14.4%

Job Involvement Level	Attrition Rate
Level 4	9.0%

Environment Satisfaction	Attrition Rate
Level 1	25.4%
Level 2	15.0%
Level 3	13.7%
Level 4	13.5%

Appendix 9: Correlation Analysis

Variable	Correlation with Attrition
Overtime	+0.246
Job Level	-0.169
Monthly Income	-0.160
Seniority	-0.134
Job Involvement	-0.130
Employee Satisfaction	-0.103

Appendix 10: Predictive Model Performance

Metric	Value
Accuracy	80.3%
AUC	0.73
Cross-Validated AUC	0.748 $\pm$ 0.013
Recall	19.1%